UGANDA CHRISTIAN UNIVERSITY
A Centre of Excellence in the Heart of Africa

FACULTY OF ENGINEERING, DESIGN AND TECHNOLOGY
DEPARTMENT OF COMPUTING AND TECHNOLOGY
EASTER 2026 SEMESTER TEST

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1. **PROGRAM**: BACHELOR OF SCIENCE IN COMPUTER SCIENCE
2. **YEAR**: 3, 2
3. **SEMESTER**: 2
4. **COURSE NAME**: SOFTWARE CONSTRUCTION

Examination Instructions

You are provided with a fully working National E-Voting console application packed into one file. It handles candidate CRUD (with eligibility checks for age, education, criminal record), voting station management, position and poll lifecycle, voter registration and verification, ballot casting with duplicate prevention, result tallying with bar charts and turnout statistics, and role-based access for four admin types plus voters.

Run it first. Log in as admin, create a station, register a voter, set up a poll end-to-end, then vote. Understand every feature before you touch the code.

The application was deliberately written to violate every principle you have studied and Your Task is to refactor this monolith into a modular, object-oriented Python project. The application must behave identically after refactoring e.g., same menus, same prompts, same outputs. Do not add new features.

Apply the following principles from this course:

1. **Modular Design**
2. **Object-Oriented Design**
3. **Separation of Concerns**
4. **Clean Code**

Deliverables & Assessment

Create a GitHub repository and push a project folder with all source files plus a brief report/ GitHub `readme.md` (1–2 pages) explaining your structure and design decisions.

Criterion	Weight	Focus
Modular Structure	25%	Logical file separation, single responsibilities
Object-Oriented Design	20%	Proper classes, encapsulation, meaningful methods
Separation of Concerns	20%	UI, logic, and data layers fully decoupled
Clean Code Quality	15%	Naming, readability, no duplication
Working Application	10%	All original features still function
Report	10%	Clear explanation of structure and decisions

◀ END ▶